Why some of Environment Canada's wind numbers are in knots

Kevin J.S. Duska Jr. · Oct. 29, 2026

Some of the Oct. 17 wind data I've been linking to is in knots, and some is in km/h. They come from different Environment and Climate Change Canada (ECCC) products. Here's what each one is, how I confirmed the units, and what the model data says once it's read correctly.

Bottom line

The wind speeds in knots come from a separate ECCC product: the **regional model forecast profiles** ("ProgTephi" files) from the Regional Deterministic Prediction System (RDPS). They're raw computer-model output for one point, the CFB Wainwright airfield (CZWN). The public forecasts and wind warnings are a different product: forecasters write them in km/h for the public.

ECCC's own documentation says the model wind speed is in knots. Read that way, all four model runs, from the one issued at 6 p.m. MDT on Oct. 15 onward, forecast the wind swinging to west-northwest between 3 and 6 p.m. on Oct. 17. That's the same thing the public forecasts said in words.

Four ECCC products, two units

The two public products are in km/h. The two technical upper-air products, the kind pilots and forecasters use, are in knots. One knot is 1.852 km/h.

Product	What it is	Wind unit	Who uses it	Oct. 17 example
Public forecast (FPCN16)	Forecaster-written text for Lloydminster – Wainwright – Vermilion – Provost	km/h	Public, media	5 a.m. forecast
Weather warning (WWCN16)	Forecaster-issued wind warning	km/h	Public, emergency managers	2:14 p.m. warning
RDPS forecast profile (ProgTephi)	Computer-model forecast for the CZWN airfield, from the surface to 10 hPa, every 3 hours out to 48 hours; runs at 00Z and 12Z	**knots**	Forecasters, aviation, specialists	12Z run, Oct. 17
Observed sounding (ObsTephi)	Weather-balloon measurements; the nearest site is Stony Plain (Edmonton), about 200 km west	**knots** (column headed KTS)	Forecasters, aviation, specialists	6 a.m. MDT balloon

The model profiles are among the guidance forecasters work from when they write the public text. I don't know whether anyone at the base saw model output. The military's own weather service (CFWOS) issues its own forecasts, and that's one of the questions I've put to DND.

What the model said for CZWN on Oct. 17

Every run had a strong west-southwest wind at 3 p.m., swinging to west-northwest (288–302°) by 6 p.m. These are the model's average winds, not gusts, in knots with km/h in brackets. Times are MDT.

Model run (issued)	3 p.m., near surface	3 p.m., 850 hPa (≈1.5 km up)	6 p.m., near surface	6 p.m., 850 hPa
6 p.m. Oct. 15 (00Z Oct. 16)	258°, 22 kt (40)	258°, 36 kt (66)	297°, 19 kt (35)	302°, 37 kt (68)
6 a.m. Oct. 16 (12Z)	264°, 23 kt (43)	265°, 39 kt (73)	299°, 18 kt (34)	299°, 35 kt (65)
6 p.m. Oct. 16 (00Z Oct. 17)	257°, 25 kt (46)	259°, 43 kt (79)	300°, 17 kt (32)	299°, 32 kt (60)
6 a.m. Oct. 17 (12Z)	249°, 22 kt (41)	252°, 38 kt (71)	288°, 19 kt (35)	289°, 33 kt (62)

- **Near surface:** the file repeats one value at every level from 1015 to 925 hPa, because those pressure levels sit at or below the airfield's ground level. That repeated value is the model's lowest-level wind.
- **Timing:** output comes every 3 hours, so the model places the shift somewhere between 3 and 6 p.m. The CZWN airfield observed it at about 3:16–3:28 p.m., with an 82 km/h gust from 269° at 3:16.
- **Gusts:** the model gives average wind only. The gust figures are in the public forecast ("west 50 gusting to 80") and the warning ("up to 100 km/h").

How I confirmed the unit, and why it matters

The forecast files don't print a unit beside the wind speed, but ECCC's documentation for them does:

> `UV : Wind speed (knots)`

It says the same for the observed soundings, whose column header reads KTS. Sources: ECCC's Datamart vertical-profile documentation for model profiles and observed profiles (ECCC-MSC/open-data, read Oct. 29).

> If you read these numbers as km/h, every model wind drops by almost half. The model would seem to call for a light breeze (22 "km/h" at 3 p.m.), when it actually forecast about 41–46 km/h near the ground and 66–79 km/h at 1.5 km. The knots reading also matches the Stony Plain balloon measurements, which are labelled in knots.

What I'll say, and what I won't

I can say:

- ECCC's regional model, in four runs from the evening of Oct. 15, forecast the wind turning west-northwest on the afternoon of Oct. 17.
- At 3 p.m., the model forecast about 41–46 km/h near the ground and 66–79 km/h at 1.5 km up. The original files give those speeds in knots.
- The public forecasts and warnings, in km/h, said the same thing in words.

I can't say:

- that anyone at CFB Wainwright saw the model output;
- that the model forecast the gusts (it gives average wind only);
- that the model timed the shift more precisely than "between 3 and 6 p.m."

Sources

- ECCC MSC Datamart, RDPS forecast profiles for CZWN: 00Z Oct. 16 · 12Z Oct. 16 · 00Z Oct. 17 · 12Z Oct. 17. Copies with SHA-256 checksums are archived in our part 2 repository.
- ECCC documentation: model profiles · observed profiles

- ECCC public forecast, 5 a.m. Oct. 17 and wind warning, 2:14 p.m.
- ECCC Stony Plain sounding, 6 a.m. MDT Oct. 17

Kevin J.S. Duska Jr., President, Prime Rogue Inc. · kevin@primerogueinc.com · 438-937-2594 · primerogueinc.com

---

Prime Rogue Inc. · Uses ECCC open data (Open Government Licence – Canada). Analysis and text CC BY-NC 4.0. ECCC's Datamart keeps files for a limited time, so if a link has expired, ask me for the archived copy.